TRUHEAL-LINK

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TRUHEAL-LINK

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Project Approval and Endorsement

"We hereby declare that we have read this project documentation and in my opinion this project is sufficient in terms of scope and quality for the award of the degree of BS in Computer Science."

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We declare that this project title entitled "**TRUHEAL-LINK**" is the result of our own research and development except as cited in the references. This project has not been accepted for any degree and is not concurrently submitted in candidate for any other degree. At any time if our statement is found to be incorrect even afterwards of BS Computer Science, the university has the right to withdraw our BS Computer Science degree.

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Acknowledgement

We are grateful to Almighty Allah for giving us the strength and the ability to complete this project titled “TruHeal-Link.”

We would like to express our sincere gratitude to our supervisor, Dr. Shahzadi Tayyaba, for her continuous guidance, valuable suggestions and support throughout this work. We are also thankful to the University of Education for providing a positive and helpful academic environment.

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Date: December 17, 2025

Abstract

TruHeal-Link is a secure and verified healthcare platform designed to address the growing problems of online medical misinformation. Many users depend on social media for health-related decisions resulting in confusion and medical risks. TruHeal-Link addresses these issues by offering verified guidance through licensed doctors, phychologists, AI tools and SOS emergency system.

Key features include doctor verified content, AI-guided responses, appointment scheduling, personalized recommendations and a secure wallet for professional payments and rewards. Mental health support is enhanced through psychologist involvement and private counseling features. Observations from the development process highlight that users feel more protected and confident when information is verified and emergency alerts operate immediately. Doctors earn money by conducting patient appointments on the platform. The patient pays first the system deducts commission and the remaining amount is added to doctor's wallet. Doctors can withdraw their earnings and also earn reward points and badges for verifying health posts.

While TruHeal-Link improves access to trustworthy medical guidance it is limited by internet availability and cannot replace in person clinical treatment. Overall, the platform aims to reduce misinformation, increase user safety and enhance both mental and physical well-being through a reliable and easy to use healthcare solution.

List of Tables

Table 2.1----------9

Table 2.2----------11

List of Figures

Figure3.1----------21

Figure 3.2----------22

Figure 3.3----------23

Figure 4.1----------25

Figure 4.2----------27

Figure 4.3----------30

Figure 4.4----------32

Figure 4.5----------33

Figure 4.6----------36

Figure 4.7----------38

Figure 4.8----------40

Table of Contents

TRUHEAL-Link----------i

Declaration----------iv

Plagiarism undertaken----------v

Acknowledgement----------vi

Abstract----------vii

List of Tables----------viii

List of Figures----------ix

Chapter 1 Gathering and Analyzing Information---------- 1

1.1 Introduction----------2
1.2 Problem Statement----------2
1.3 Goals and Objectives----------3
1.4 Research Questions----------3
1.5 Methodology----------4
1.5.1 Available Methodologies----------4
1.5.2 Chosen Methodology----------4
1.5.3 Reasons for chosen Methodology----------4
1.6 Definitions, Acronyms and Abbreviations----------4

Chapter 2 Software Requirements Specification----------6

2.1 Literature Review----------7

2.2 Stakeholders Characteristics----------7

2.3 Domain Requirements----------8

2.4 Functional Requirements----------9

2.5 Non-Functional Requirements----------11

Chapter 3 Analysis----------13

3.1 Use Cases----------14

Chapter 4 Design----------24

4.1 Architecture Diagram----------25

4.2 ERD----------27

4.3 Data Flow Diagram----------30

4.5 Class Diagram----------33

4.6 System Sequence Diagrams----------36

Chapter 1

GATHERING AND ANALYZING INFORMATION

1.1 Introduction

Healthcare is a basic need for everyone but many people still face problems finding accurate and correct medical information online. People commonly rely on online searches or social media when someone feels sick, they often check online post or videos to find quick help. The problem is that the majority of this content is false or not verified by doctors and can sometimes result in panic or wrong decisions. Therefore, users cannot easily decide which information is safe to trust and which is risky.

For many people, visiting a doctor is not simple or convenient. People who stay away from hospitals, have busy schedules or lacking financial resources often cannot receive medical help on time. Because of this many individuals ignore small symptoms or depend on incorrect online tips causing minor health issues to become serious.

To address these problems, we created TruHeal-Link a secure healthcare platform that offers AI guidance along with emergency situations. The system connects people with qualified doctors and phychologists who review and verify all health-related content ensuring that users always receive correct and reliable device. Every doctor and post is verified to ensure authenticity and safety. Users can consult with verified doctors, access health remedies whether it is tips on managing a cold, guidance on healthy eating or advice for mental wellness and get quick responses to health-related questions from AI. The AI also enhances user experience by recommending suitable doctors, posts and relevant friends based on each user's profile.

The platform also includes an SOS alert feature that can quickly notify doctors or trusted contacts during an emergency. TruHeal Link brings everything together-verified health content, professional advice, AI support and emergency help to make healthcare simple and trustworthy.

1.2 Problem Statement

As digital health platforms grow, more people depend on the internet for medical guidance. But much of the online information is unverified and social media often spreads incorrect or misleading health tips. Because of this, many users cannot separate real advice from false information putting them at risk of making poor health choices. Even minor health problems can become serious if they remain unnoticed.

Offline doctor visits are usually expensive and not easily available for those who live far away or have fewer resources. In emergencies this becomes more dangerous because people cannot quickly inform their doctors or loved ones. Any delay or wrong advice can be extremely risky. These issues show that the current healthcare system does not give people the immediate support they need.

The situation highlights the need for a reliable healthcare platform that includes verified doctors, correct medical details, AI support and quick emergency notifications. With all these features integrated, users can receive accurate advice, immediate help and personalized suggestions. It allows people to make safer health decisions and reduces the spread of wrong information. Such a platform can increase trust and support both mental and physical health for all users.

1.3 Goals and Objectives

The main purpose of this project is to create a reliable healthcare app where users can find correct medical information and talk to verified doctors.

- All health posts must be checked by relevant doctors. This objective makes sure that the information shared is correct and trustworthy and avoids misinformation that could harm users.

- Allow users to share their health issues and get trusted advice. People can ask questions and receive responses from trained professionals instead of relying on random internet content.

- Introduce subscription-based income option for doctors. Doctor can subscribe to monthly plans after a free trial and earn through the platform.

- Use AI and SOS features to provide fast responses during urgent situations. AI will reply to common health questions instantly. In emergencies the SOS option sends instant alerts to selected contacts or doctors reducing delays in critical moments.

- Encourage doctor participation through reward and point systems. The more a doctor engages the more points and reward they receive. This keeps them motivated to stay involved.

- Highlights doctor performance with a ranking or level badge. The platform will assign each doctor a level or badge based on how active and helpful they are. Doctors who verify more posts, responds quickly and maintain good interaction with users will influence their ranking. This helps users easily recognize the most helpful and trusted doctors.

- Support all financial operations with a safe-in app wallet system. All payments, rewards and subscriptions are handed through a secure wallet. This makes money transfer easy and safe.

1.4 Research Questions

1. How can this platform help stop the spread of wrong health information online?
2. Will user feel safer and more confident when posts are doctor verified?
3. How well can the AI assistant guide users with basic health questions?
4. How fast can the SOS feature respond during health emergencies?
5. How do reward and subscriptions motivate doctors to participate?
6. How does the level badge show which doctor is more reliable?
7. How does adding phychologists improve support for mental health users?

1.5 Methodology

1.5.1 Available Methodologies

There are different methodologies used for software development. One option is the Waterfall Model which follows a step-by-step process where each phase must be fully completed before moving to the next. Another commonly used approach is the Agile Model in which the project is created in small sections, allowing updates or changes whenever needed. The iterative model improves the software with every cycle while the Spiral model looks for possible risks and problems as the project moves forward. Prototype Model makes a small sample first so users can see how the final software will look.

1.5.2 Chosen Methodology

This project uses the Agile Model. Work divides into small sprints with each sprint focusing on a single feature like wallet management, AI assistant, emergency SOS or the reward system for doctors.

1.5.3 Reasons for Chosen Methodology

Agile is ideal because the project has many features that may change based on testing and user feedback. It allows teams to work on multiple parts at the same time encourages regular communication and ensures continuous testing. This helps maintain accuracy which is very important for a healthcare system.

1.6 Definitions, Acronyms and abbreviations

- **Health Misinformation:** Any medical claim or advice that is not verified and may mislead a patient.
- **Verified Doctor:** A doctor whose qualification and license have been checked and approved by the platform.
- **AI-Artificial Intelligence:** Automated intelligence used to deliver quick responses and recommendations.
- **SOS-Emergency Alert**: A built-in feature designed to notify emergency contacts or available doctors instantly providing immediate assistance during critical situations.
- **User**: A registered person using the app for health guidance, AI assistance or consultations.
- **Admin-Administrator**: The person responsible for managing the system, verifying doctors and handling reports.
- **Subscription Model**: A monthly plan that allows doctors to access paid features and earn through the platform.
- **In-App Wallet**: A secure digital wallet for handling financial transactions inside the app.

- **Reward System**: A performance-based point mechanism for doctors.
- **Doctor Badge/ Mentor Level**: A ranking symbol showing the doctor's performance and contribution level.
- **Medical Post Verification**: The checking and approval of user posted health information by licensed professionals.
- **Phychologist**: A mental health helper who supports users dealing with stress and emotions.

Chapter 2
SOFTWARE REQUIREMENTS SPECIFICATION

2.1 Literature Review

The spread of wrong health information on social media is a serious public health problem. Suarez-Lledo and Alvarez-Galvzez (2021) studied 69 research articles to measure how much false health information exists online. They found that false information about smoking and drugs was extremely high and vaccine misinformation was also common. Most misleading content was about diseases, vaccine and treatments. Twitter had the highest amount of false health information followed by Facebook and YouTube. The study highlighted that the lack of verified medical content on social media allows harmful information to mislead content (Suarez-Lledo & Alvarez-Galvez, 2021).

Recent advances in AI-powered telemedicine have greatly improved healthcare services by making them more accessible and accurate especially for people living in remote areas. Sharma et al. (2023) analyzed how AI can be effectively used in telemedicine platforms. Their findings showed that AI improves patient monitoring, diagnosis and clinical decision support. During the COVID-19 pandemic, telemedicine use increased around the world. AI tools can study medical images, patient records and gives basic health advice through virtual assistants. Still challenges like weak technology, data safety problems and training needs must be solved (Sharma et al., 2023)

The use of AI with mobile health applications has created new ways to improve patient care and health monitoring. Haleem et al. (2021) reviewed how telemedicine works in modern healthcare systems. Their study showed that AI supports real time health monitoring and help doctors make faster decisions. Mobile apps and wearable devices can track health signs such as heart rate and physical activity. It also helps patients in remote areas and reduce the need to visit hospitals. Still, problems such as privacy risks, technology barriers and weak patient-doctor connection were highlighted (Haleem et al., 2021)

Emergency alert systems in mobile health applications are very important for patient safety and fast medical help. Isong et al. (2022) developed the “Help Pro” SOS app for android to assist people in urgent situations. The study focused on rural South Africa where slow ambulances and poor roads caused preventable deaths. The app uses GPS and Google Maps to find the nearest ambulance and handle multiple requests using a First Come First Served system. Features include one-tap alerts, location sharing, emergency contacts and blood donation requests. The app reduces delays but needs good networks and regular use (Isong et al., 2022).

2.2 Stakeholders Characteristics

1. End Users (Patients / General Users): These are people who use the application to search for verified health information, consult doctors, ask questions and access the AI assistant. They may come from different age groups and expect information that is dependable, simple and easily accessible.

2. Medical Doctors (Certifies Professionals): Doctors are licensed professionals who register on the platform to offer consultations**.** Doctors verify posts, treat users online and provide expert suggestions. They want a system where their identity is protected, only verified doctors can join and the interface is simple for checking content and handling appointments.

3. Mental Health Professionals: These professionals work with users who need support or emotional help. They check mental health related posts and require a private secure place to talk with patients.

4.**System Administrator**: Admins handle doctor verification, manage user activities, monitor reports and ensure platform safety. They manage revenue, subscription plans, and financial operations through the wallet system.

5.**Emergency Contacts**: These are trusted contacts chosen by users to receive SOS alerts during urgent health situations. They expect timely alerts, accurate tracking and clear notification message.

6. **Project Investors / Owners**: They oversee the financial and operational growth of the platform. They expect user satisfaction, strong performances and a stable revenue model.

2.3 Domain Requirements

Domain requirements are the rules, processes and limitations that come directly from the specific field in which the software will be used. These requirements come from real world applications and industry practices that cannot be changed or ignored. They explain how the system must behave according to the standards, laws and workflows of that domain.

The system must allow only verified medical professionals to approve or share medical content. Every health-related post must be reviewed by relevant doctor (e.g. skin posts by dermatologists' mental health posts by phychologists). Domain rules strictly prohibit unverified health suggestion. The system should support multiple roles, including users, doctors, phychologists and administrators. Strong privacy protections are essential because users share sensitive data like medical conditions, diet and stress levels. Accurate information is vital to prevent health hazards. Emergency functions such as SOS alerts and location sharing must respond instantly. Subscription plans, wallets and reward points should be managed securely.AI can assist with general health inquiries but should not replace verified doctor guidance for serious medical situations.

2.4 Functional Requirements

Category	Functional Requirements
User Module Requirement:	• The app must allow users to create an account using their professional details like age, name, gender. • Users can optionally add information related to health habits such sleeping patterns, stress or diet. • The system must show relevant posts and doctor recommendations based on user data. • When a user uploads a post, the system should send it to the relevant doctors based on post type (e.g. skin, hair, fitness). These specialists will review and verify the post before it becomes public. • Users use the AI assistant for quick help with common health doubts. • Appointment booking with verified doctors must be available. • All financial transactions should be processed through in app wallet system. • Emergency contact information must be collected from every user.
Doctor Module Requirements:	• Doctors must be able to register by submitting their qualification, license and medical background. • Only after approval from the admin team should doctors gain access to the doctor specific features. • Doctors should have the option to verify or reject user posts. • Doctors must earn reward points when verified posts receive a higher number of likes (e.g. more than 1000 likes). • Doctors must have the access to chat or appointment features to communicate with patients. • A subscription system with trial access should be provided.
Phychologist Module Requirements	• Phychologists should be able to create an account by entering their qualifications and license for verification.

	• The admin must approve the phychologist before they can access the system. • Phychologists must review mental health related posts and verify them. • They must be able to provide chat based or booking counseling support.
Emergency Module Requirements:	• The system must store emergency contact details for each user. • The SOS system must be activated using a simple gesture or button. • Alerts must include user location and be sent to the emergency contact or doctor. • Emergency notification must reach recipients without delay.
Admin Dashboard Requirements:	• The admin must have the ability to verify doctors or phychologists before giving them access. • Admin must view all health posts and take action on harmful content. • Admin must manage user reports or ban accounts if needed. • The admin dashboard should support managing accounts, subscriptions commissions and doctor rewards. • The admin must be able to monitor doctor performance, activity levels and mentor status. • Admin must oversee the overall health of the platform.

Table 2.1

2.5 Non- Functional Requirements

These describe how well the system should work.

Category	Non- Functional Requirements
Performance Requirements:	• The app must open within 3-5 seconds on normal internet speed. • AI generated answers should appear in 2 seconds. • SOS notifications should be delivered instantly without waiting time.
Security Requirements:	• All user data must be encrypted. • Secure login should include OTP or two factor verification. • Only authenticated and approved doctors can view or post medical content. • Wallet operations must follow strict financial protection rules.
Usability Requirements:	• Interface must be simple, clean and easy for non-technical users. • Buttons and menus must be easy to navigate. • The AI chat must communicate in a friendly, helpful and natural way.
Reliability Requirements:	• System should stay available 99% of the time. • SOS services must work even with weak network signals. • Backup and recovery tools must safely store all important data.
Maintainability Requirements:	• Code should be organized and structured so updates are simple. • Adding upcoming features like video calling and smart wearable monitoring should integrate smoothly.
Compatibility Requirements:	• The application must support both android and iOS platforms. • It should display properly on all screen sizes and resolution.

Ethical Requirements	• The platform must remove all dangerous or misleading medical content. • AI must not give decisions that replace a real doctor's advice. • User privacy especially medical data must be protected.

Table 2.2

CHAPTER 3

ANALYSIS

3.1 USE CASES

UC Number: 1

UC Name: User Registration

Primary Actor: Patient (User)

Secondary Actor: Admin

Description: This use case covers how a new user signs up on TruHeal-Link. The user provides essential personal information which the system verifies before granting access to health-related features.

Pre-Conditions:

User must have a valid email or phone number.

User must agree to platform policies.

Account must not exist already.

Main Success Scenario:

- Users open the app interface
- Chooses Create Account
- User fills required information
- System validates data
- Optional health info added
- Account successful created
- Confirmation message appears

Alternative Scenario:

Contact invalid ->Error shown

Password too simple ->System suggests stronger one

Contact already used ->Registration blocked

Post Conditions:

Account saved in system database.

User can now log in.

Extensions:

OTP Verifications:

- OTP sent to user.
- User enters code.
- Correct OTP ->Account activated.
- Wrong OTP -> System requests retry.

UC Number: 2

UC Name: Professional Registration

Primary Actor: Doctor, Phychologist

Secondary Actor: Admin

Description: TruHeal-Link allows licensed doctors and phychologists to register by providing detailed information and verification documents. Their account becomes active only after Admin approval.

Preconditions:

Applicant must have authentic and valid licenses.

Information provided must be correct.

Main Success Scenario:

- Doctor/Psychologists opens the app
- Selects Register as Professional
- Enters personal and professional details
- Uploads legal documents and certificates
- System saves profile and sends it for Admin review.
- Admin examines all documents
- Admin approves the application
- System activates professional features

Alternative Scenario:

Missing documents -> System requests complete uploads

Invalid / expiry license -> Admin rejects application

Blurry or unclear documents -> Admin requests re-upload

Postconditions:

Professional account activated.

Specialists can verify posts and offer consultations.

Extensions:

Admin may ask for additional verification steps if needed.

UC Number: 3

UC Name: Admin Platform Management

Primary Actor: Admin

Secondary Actor: System, Doctors, Users

Description: Admin verifies doctors/phychologists, manages reports, removes harmful content, bans users if needed, handles wallet, subscriptions and platform activities.

Preconditions:

- Admin is logged in
- Verification requests or reports exist in the system.

Main success scenario:

- Admin opens dashboard.
- Checks doctors/psychologists' verification requests.
- Approves or rejects based on documents.
- Review user reports.
- Removes fake/ harmful posts.
- Warns or bans users/ doctors if required.
- Manages wallet, payments and subscriptions.
- Monitors all platform activity.

Alternate Scenario:

If verification fails -> admin rejects.

If payment issue occurs ->admin reviews logs

Harmful content -> admin deletes or warns user.

Post Conditions:

All admin actions such as verifications, content removal are saved in the system. Wallet and payment records are updated to keep the platform safe and properly managed.

UC Number: 4

UC Name: User Health Post Creation

Primary Actor: User

Secondary Actors: Doctor, Psychologist

Description: This use case explains how users are allowed to share information or query-based posts regarding health issues. Before being published, every post is checked by a certified specialist to maintain accuracy and safety.

Pre-Conditions:

- User must be signed into their account.
- Uploaded Content must follow the platform rules.

Main Success Scenario:

- User clicks "New Post".
- User Provide text or media content.
- User also identify the Post type and category i.e. it related to skin or hair etc.
- System sends the post to a qualified expert doctor or psychologist for Review.

- System sets the post badge as “Pending Verification”.

Alternative Scenario:

- Irrelevant Post –> Rejected by the Admin.
- Wrong media format -> user asked to replace the file.
- Post interrupted due to connectivity –> saved in drafts.

Post Conditions: Post is successfully stored and awaits approval by a specialist doctor or psychologist.

UC Number: 5

UC Name: Health Post Verification

Primary Actor: Doctor, Psychologist

Secondary Actors: Admin

Description: Doctors and Psychologists verify posts submitted by users to ensure medical correctness and authenticity.

Pre-Conditions:

- Professional is verified and logged in.
- Post assigned to the correct domain.

Main Success Scenario:

- Open pending posts queue.
- Review post details and media.
- Approve or reject posts.
- Posts that are approved show the approved badge with the Doctor or psychologist name.
- User receives notification.

Alternative Scenario:

- Misleading content -> Rejected
- Irrelevant post -> Rejected

Post-Conditions:

- Public sees the post with the verification badge that includes the doctor or psychologist name who verify the post.
- Doctors gain points or rewards for verified contributions.

UC Number 6:

UC Name: AI Health Guidance

Primary Actor: User

Description: Users receive instant responses and health advice through an AI assistant which also recommends verified doctors and related posts.

Pre-Condition:

- User is logged into the app
- AI service is active.

Main Success Scenario:

- User opens AI chat interface
- User type their health query
- AI processes the request
- System generates a response almost immediately
- Additional recommendations such as relevant doctors or posts are displayed

Alternate Scenario:

Out-of-scope questions ->AI informs the user it only handles health related queries and prompts for a relevant question.

Emergency detected -> SOS advice or alert suggested.

Post-Conditions:

Users get immediate health guidance and recommendations.

UC Number 7:

UC Name: Book Appointment

Primary Actor: User

Secondary Actor: Doctor, Phycologist

Description: Users can schedule appointments with verified doctors or psychologists through the app

Pre-Condition:

- User logged in
- Wallet balance sufficient
- Doctor must have available time slots

Main Success Scenario:

- User selects a verified professional.
- System shows available slots.
- User selects preferred date and time
- System deducts consultation fee.
- Appointment confirmed; notifications sent to user and professional.

Alternate Scenario:

- Slot taken -> choose another slot
- Insufficient balance -> add funds
- Professional unavailable -> Suggest alternatives

Post Conditions: Appointment recorded and confirmed.

UC Number 8:

UC Name: SOS Emergency Alert

Primary Actor: User

Secondary Actor: Emergency Contact, Doctor

Description: User can send emergency alerts with location to contacts and doctors.

Pre-Condition:

- User logged in
- Emergency contacts added
- GPS enabled

Main Success Scenario:

- Press SOS button
- System share's location
- Alert sent to contacts / doctors
- Notifications received

Alternate Scenario:

- Network weak ->retry
- GPS off -> Prompt to enable
- Contact unavailable ->Next contact notified

Post Conditions:

Alert sent successfully.

UC Number 9:

UC Name: Wallet and subscription management

Primary Actor: Doctor

Secondary Actor: Admin

Description:

Professionals handle funds, rewards, and subscriptions within wallet.

Pre-Condition:

- Doctor logged in
- Wallet balance available

Main Success Scenario:

- Doctor opens wallet section
- System displays balance and subscription info.
- Doctor chooses an action (withdraw/renew)
- System deducts or transfer's amount
- Success notification sent

Alternate Scenario:

- Payment failed ->Retry
- Reward points ->below limit
- Subscription expired ->Forced renewal

Post Conditions:

Updated wallet and subscription status.

Use Case Models

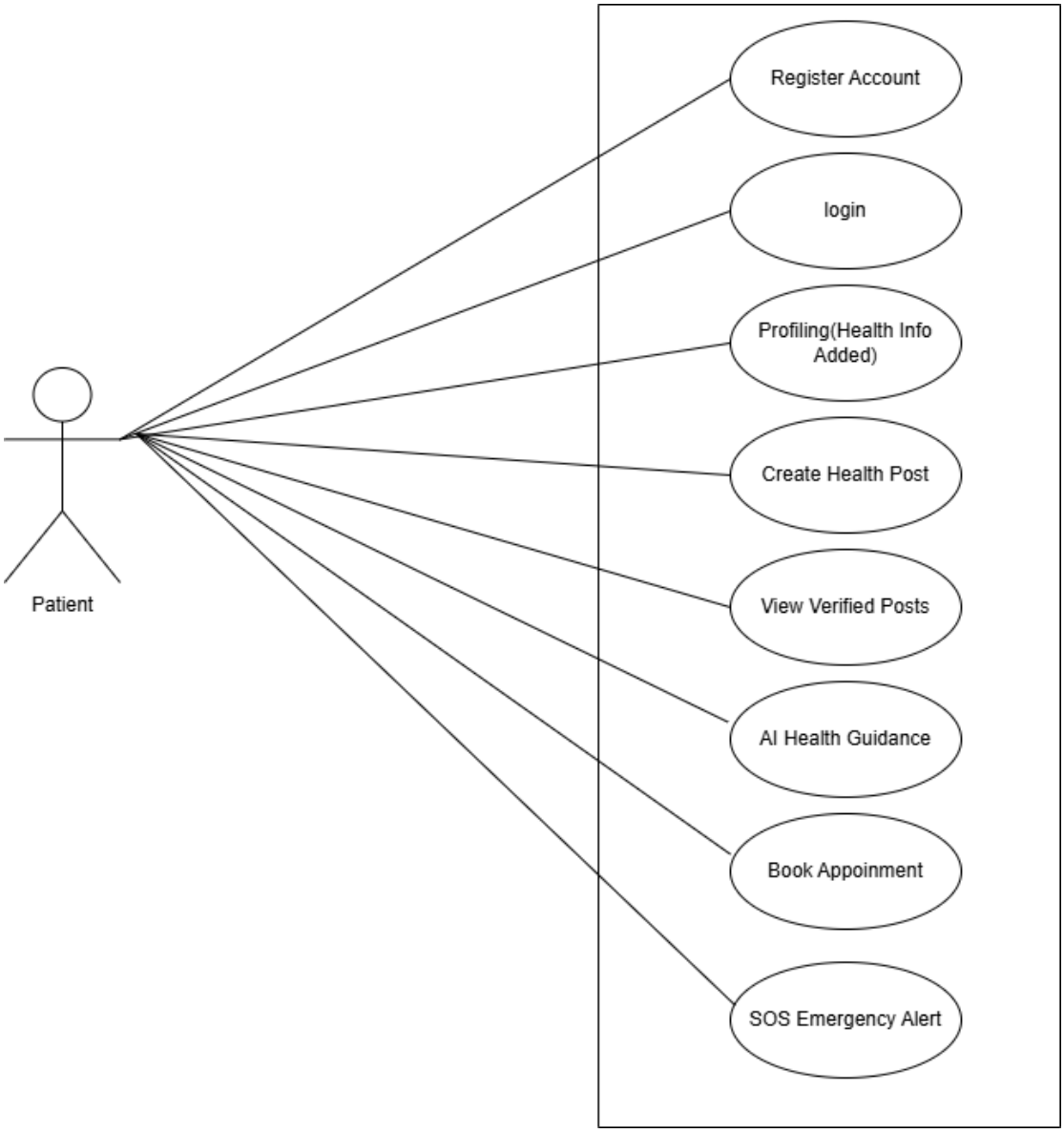

Figure 3.1: Use Case (Patient)

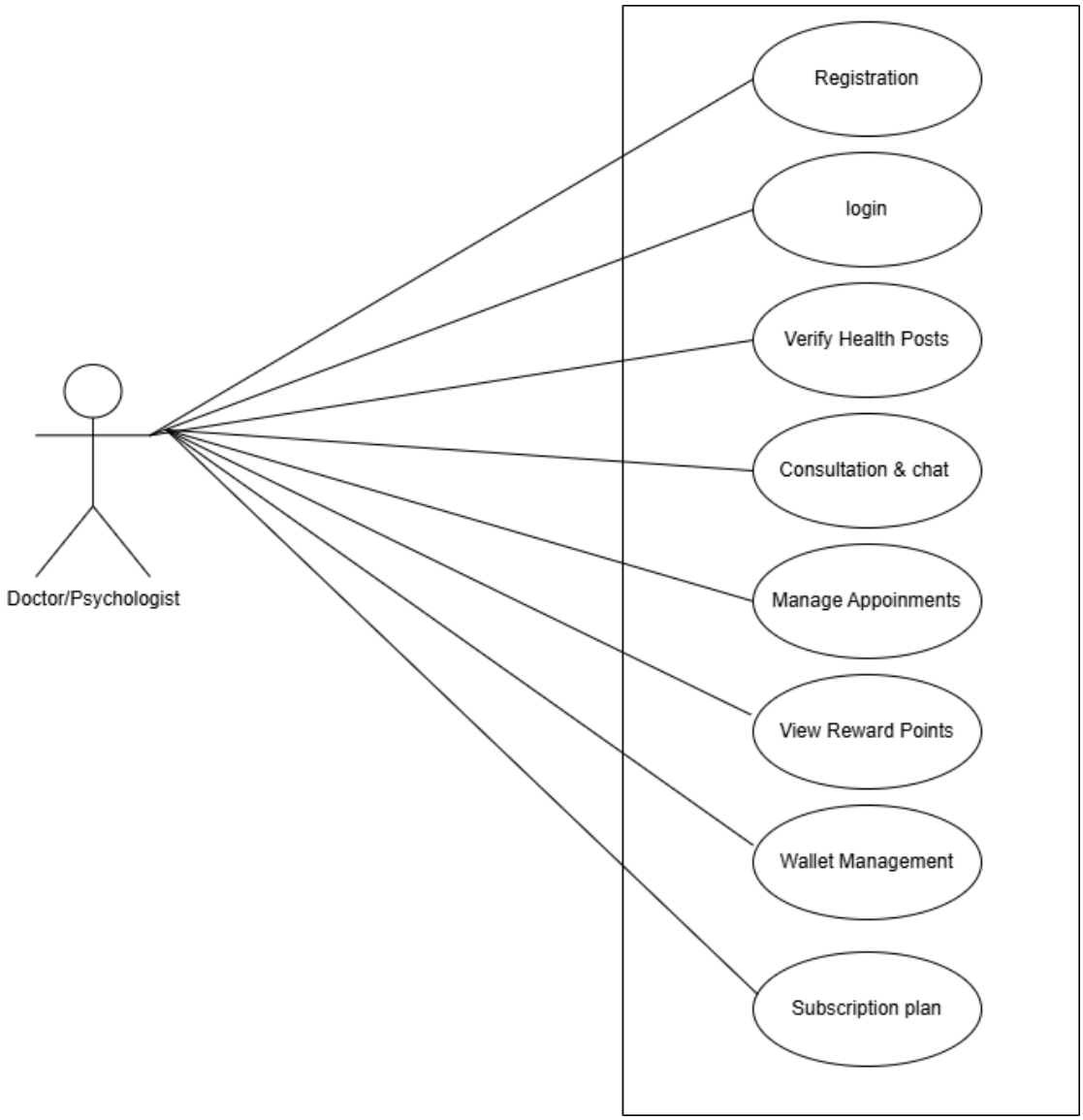

Figure 3.2: Use Case (Doctor/ Phychologist)

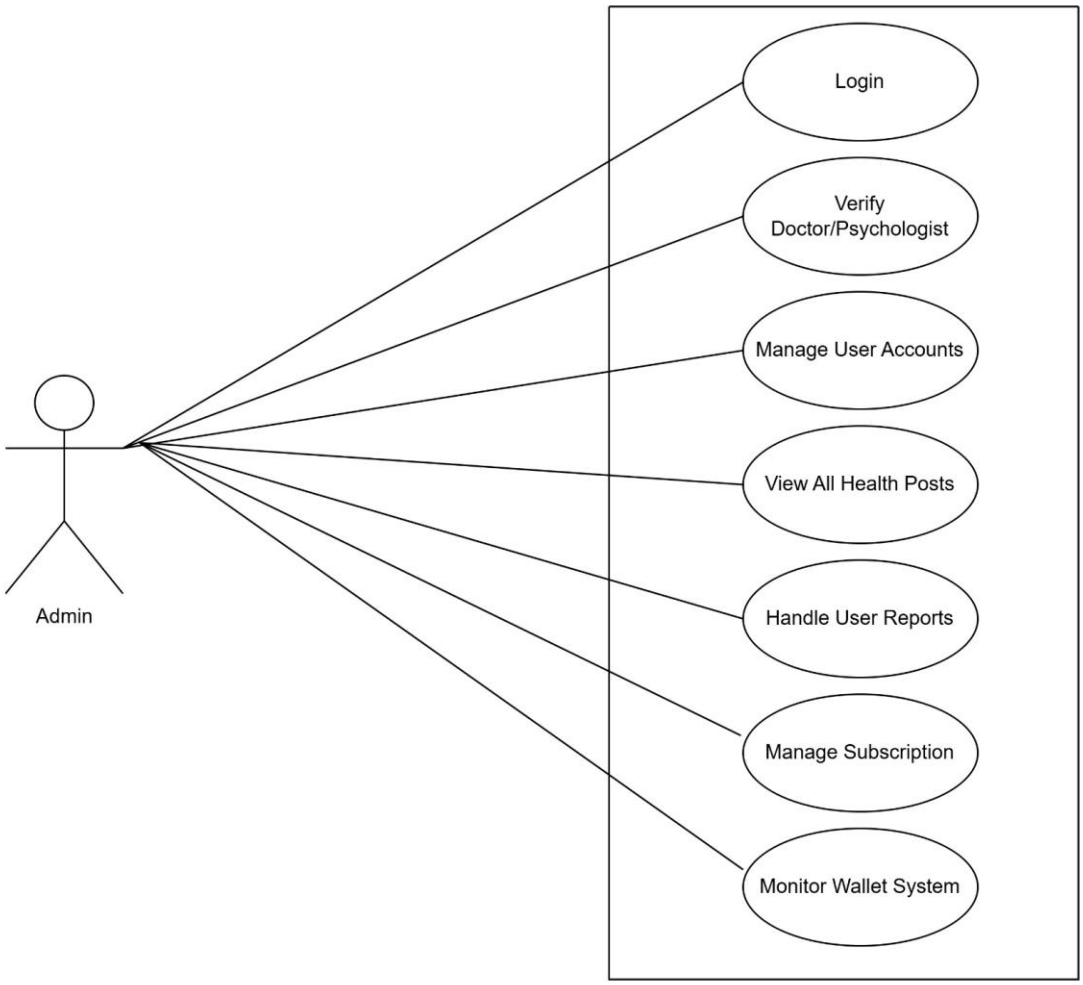

Figure 3.3: Use Case (Admin)

CHAPTER NO 4

Design

4.1 Architecture Diagram

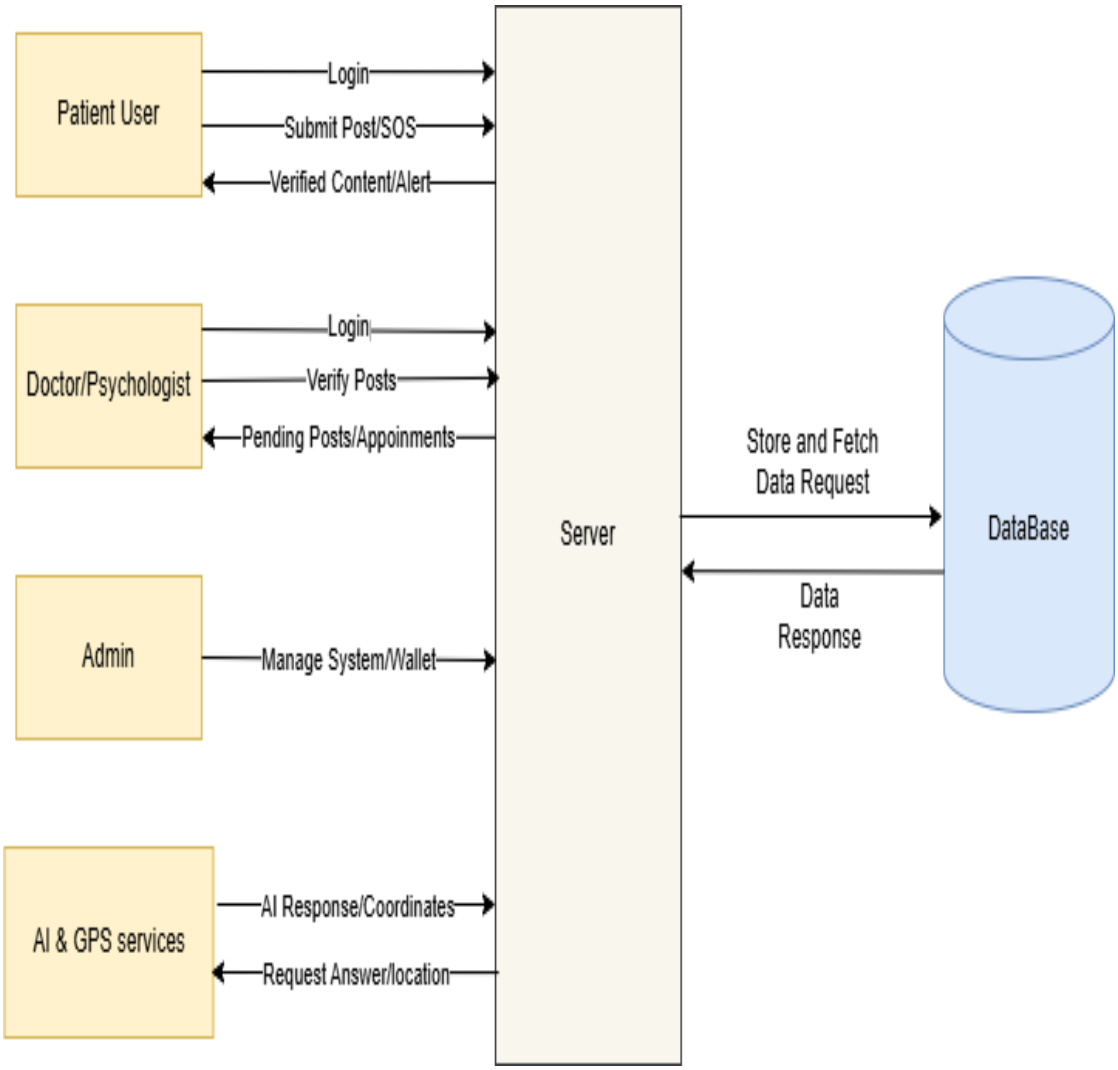

Figure 4.1 Architecture Diagram

This architecture shows data flow between users, servers, and databases in the healthcare system.

1. **User Interaction Layer**
   - Users' login and access health posts.
   - Doctors check and approve posts.
   - Phychologists handle mental health content and counseling.
   - Admin manages the platform, verifies staff, handle reports and manages payments.

2. **Server Layer**

- Checks login details
- Stores and retrieves data from the database
- Sends posts to the right specialist for approval.
- AI gives health answers and doctor recommendations.
- Sends SOS alerts and manages wallet and subscriptions.

3. **Database Layer**

- Stores information about users, doctors, phychologists posts and wallet data.
- Save appointments, contacts, transactions, and AI charts.

4. **Emergency Module**

- Stores emergency contact details.
- Send fast SOS location alerts to contacts/doctors.

Key Functional Flow

- **Authentication**: Server validates login for all user roles.
- **Post Submission**: Users submit posts; Server stores them in the Databases.
- **Post Review:** Doctors and Phychologists verify posts and update records**.**
- **Content Delivery**: Server sends approved posts back to Users**.**
- **Admin Functions:** Admin handles verification and platform activities via the Server.
- **SOS Handling:** Server sends quick alerts by reading emergency contacts from the Database.
- **Secure Communication:** Server handles all requests and responses between users and Databases, ensuring secure dataflow across the platform.

4.2 ERD

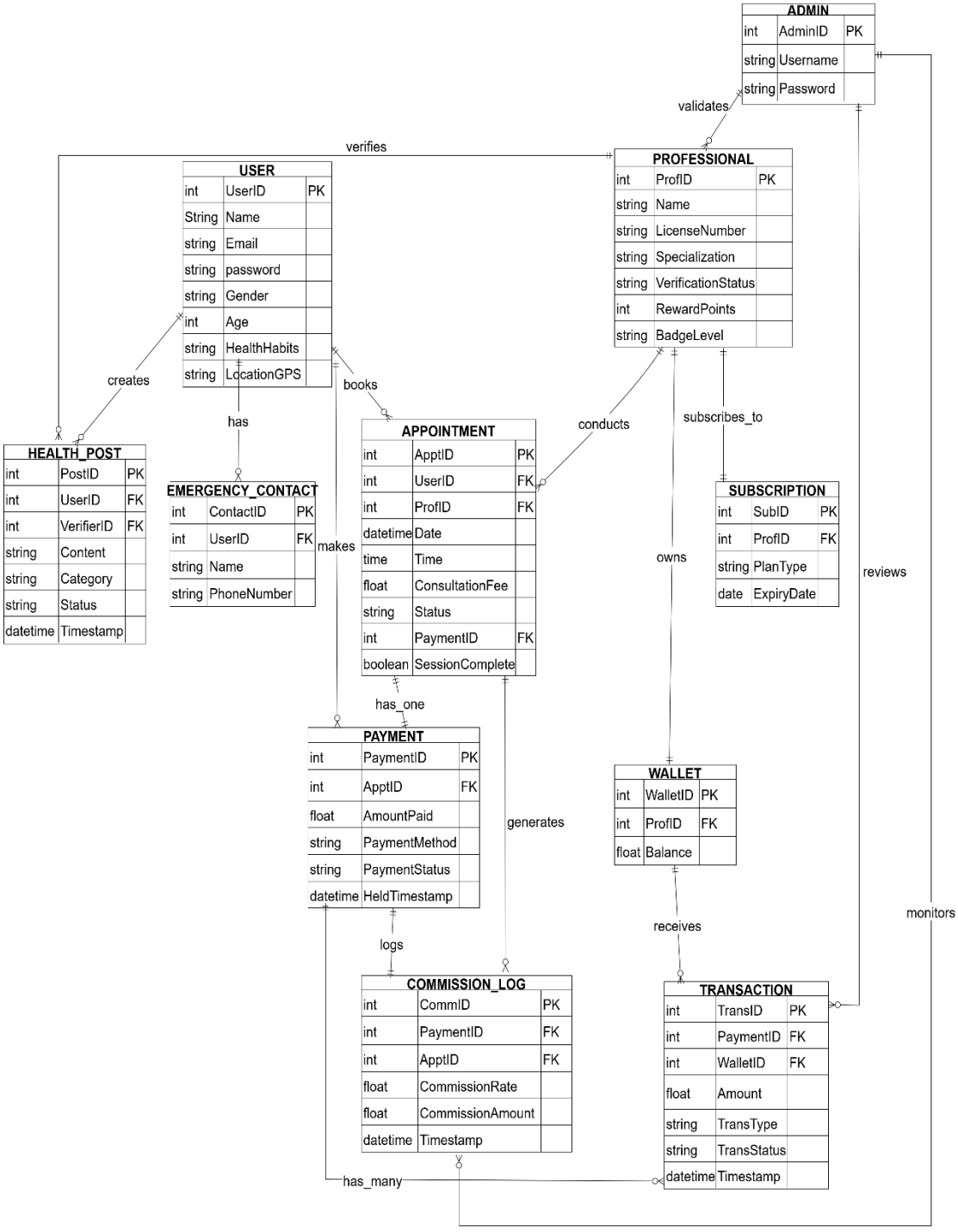

Figure 4.2 Entity Relation Diagram

The ERD illustrates the data model of the TruHeal-Link platform by defining core entities and their associations.

Entities and Attributes

- **User (Patient):** Stores patient details such as User-Id, name, email, password age, gender, health details and location. Users create accounts, add health info and emergency contacts, post health content, book appointments and pay fees. GPS location supports nearby services.
- **Professional (Doctor/Phychologist):** Includes Pro-ID, name, License Number specialization, verification-status, reward points, Badge Level. Professionals register with verified licenses. They can verify posts, conduct appointments, earn reward points, manage wallet payments and use subscriptions.
- **Admin:** Holds administrative details include Admin-ID, Username and password. Admin manage professional verification, track payments and commissions and monitor transactions to prevent fraud.
- **Emergency Contact:** Linked to the user containing emergency contact details like contact id, UserID, name, and phone.
- **Health Post:** Contains posts created by users including Post- ID, User-ID, Verifier-ID, Content, category, Time-stamp and status. Users share health posts, verified by professionals. Emergency contacts ensure user safety during health issues.
- **Appointment:** Stores appointment information such as Appointment-ID, UserID, Prof-ID, date, time, consultation fee, status, payment-ID, session complete. Appointments are booked and fees are paid in advance. Payments are held escrow until sessions are completed. Then commission is calculated, logged and the remaining amount goes to the professional's wallet. All transactions are tracked.
- **Wallet:** Stores wallet details of professionals such as Wallet-ID, Pro-ID and balance history. Wallets manage professionals' earnings. Professionals receive payments from completed appointments.
- **Subscription:** Records subscription information such as Sub-ID, prof-ID, Plan-Type, Expiry Date. Allows professionals access to platform features based on their plan.
- **Payment:** Tracks user payments for appointments including Payment-ID, Appointment-ID, amount paid, payment method, status, and held timestamp. Payments are held in escrow until session completion
- **Transaction:** Records the transfer of funds from payments to professional wallets, including Trans-ID, Payment-ID, Wallet-ID, amount, transaction type, status, and timestamp. Ensures full financial tracking.
- **Commission Log:** Tracks platform earnings for each appointment including Comm-ID, Payment-ID, Appointment-ID, commission rate, commission amount, and timestamp. Allows transparent revenue management.

Relationships

- A user creates health posts (1:M)
- A user books appointment(1:M)
- A user has emergency contacts(1:M)
- A user makes payment(1:M)
- A professional verifies health posts(1:M)
- A professional conducts appointment(1:M)
- A professional owns a wallet (1:1)
- A professional subscribes to a subscription plan (1:1)
- An admin validates professionals(1:M)
- An admin monitors commission log(1:M)
- An admin reviews transaction(1:M)
- An appointment has one payment (1:1)
- An appointment generates commission logs (1:M)
- A payment has many transactions (1:M)
- A payment logs one commission (1:1)
- A wallet receives many transactions(1:M)

4.3 Data Flow Diagram

(Level 0)

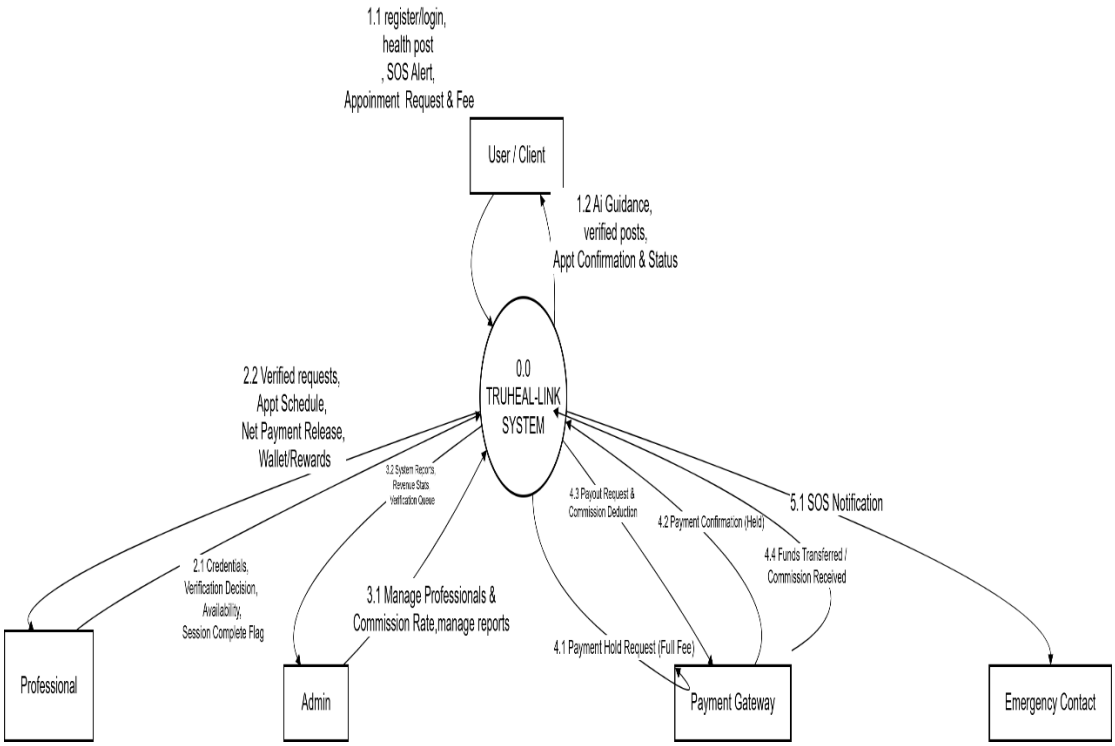

Figure 4.3 Level 0 diagram

This diagram shows the main actors and what actions they can perform inside.

Actors

- User (Patient)
- Professional (Doctor/Phychologist)
- Admin
- Payment Gateway
- Emergency contact

Flow Description:

User =>TruHeal System

- The user signs up or login to use the app
- The user shares health questions or updates.
- The user sends an emergency alert when needed.
- The user books a doctor's appointment.
- The user gets help or advice from AI.
- User pays consultation fees through the system.

TruHeal System =>User

- System provides AI guidance and verified health posts.
- After booking the system sends confirmation (time, doctor, instructions) back to the user.

TruHeal System => Professional

The platform sends:

- Verification requests to professionals who want to be verified.
- Appointment schedules so professionals know their bookings.
- System updates wallet balance, rewards and net payments after commission deduction.

Professional => TruHeal System

- Professionals send their credentials, availability and complete consultation sessions.
- They receive the systems decision about verification (approved or not).
- Professionals request payout release through the wallet system after session completion.

TruHeal System => Emergency Contact

- When an SOS alert is triggered the system sends notifications and the user's location to the emergency contact.

TruHeal System =>Admin

Admin actions the system supports:

- Approve or reject professional verification
- Ban or suspend problematic users
- Manage platform wallets and payments

Admin ->TruHeal System

- Admin also receives or reviews platform reports and statistics and processes the queue of professional verifications.
- Admin sets commission rates and monitors platform activities.

TruHeal-Link System ⇔ Payment Gateway

- System sends full consultation fee hold request to the payment gateway.
- Payment is confirmed after appointment completion.
- Commission is deducted and remaining funds are transferred to professionals.

(Level 1)

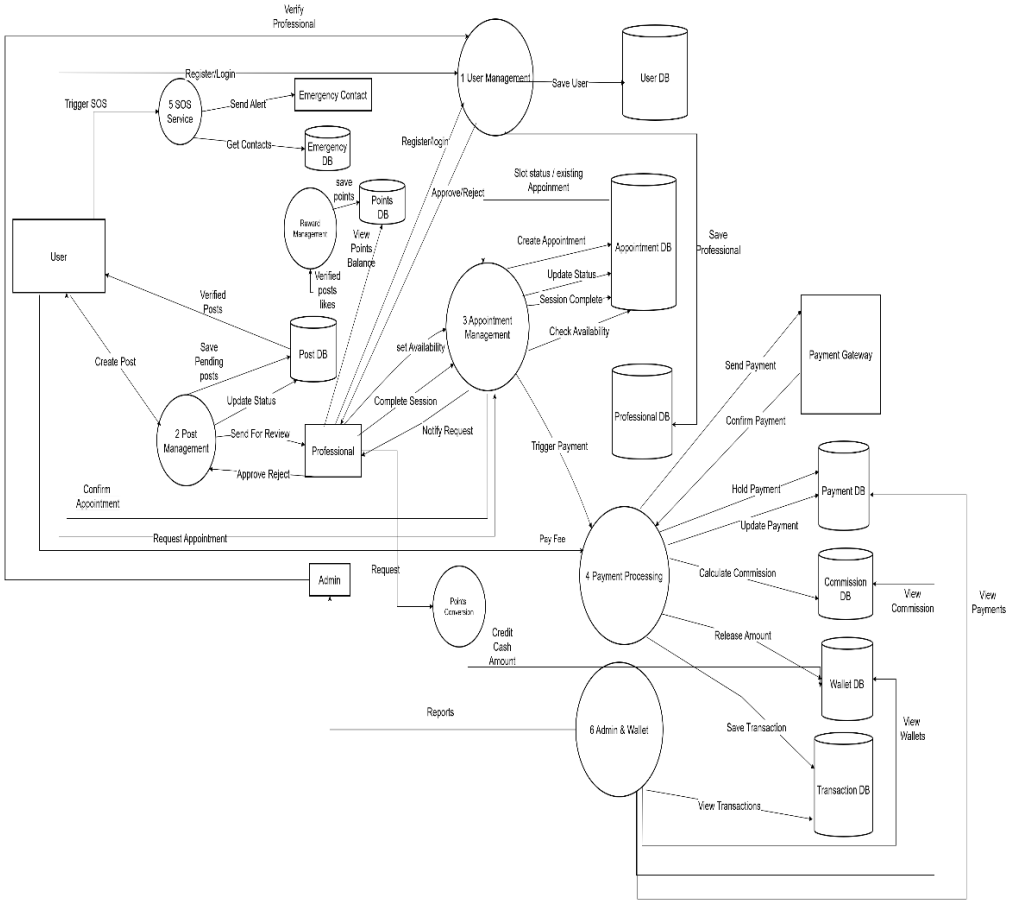

Figure 4.4 Level 1 Diagram

This diagram shows how all parts of the health system work together.

- The user can sign up or log in, create health posts, book appointments with professionals, confirm appointments and activate the SOS feature during emergencies. User related information is stored in the User Database.
- The professional receives appointment requests from users, reviews and verifies health posts, sets availability, provide services by completes sessions. Professional details and availability are stored in Professional Database.
- The SOS Service allows users to trigger emergency alerts. It retrieves emergency contacts from the Emergency Database, sends alerts to contacts and stores emergency details.
- Post Management allows users to create posts and sent for review where they are approved or rejected professionals. Post data is stored in the Post Database.
- Appointment Management manages booking, confirmation, completion of sessions, status update and professional availability. All appointment related data is saved in the Appointment Database.
- User Management manages user registration, login, profile information and professional verification. User data is stored in User Database.

- Payment Processing handles online payments through a payment gateway, manages payment confirmation, calculates commission, and updates payment records. Payment information is stored in the Payment Database and commission details are stored in the Commission Database.
- Admin and Wallet Module allow the admin to monitor the system activity, handle transactions, wallets, reports and release payments to professionals. Wallet balances are stored in the Wallet Database and all payment activities are recorded in the Transaction Database.
- Reward Management gives points to professionals for verified post likes, saves them in the points DB, allows professionals to view their points and converts points into cash credited to the wallet DB when requested.

4.5 Class Diagram

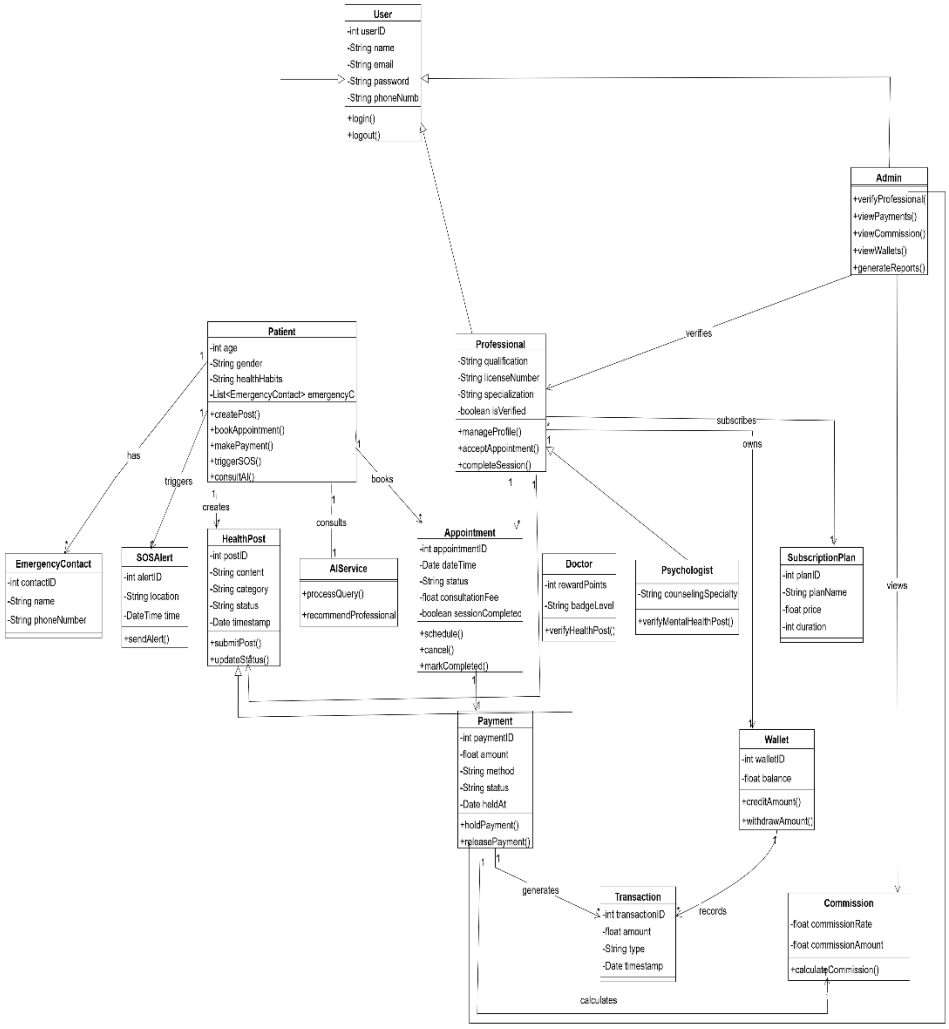

Figure 4.5 Class Diagram

This is a UML class diagram for a healthcare management system.

User Hierarchy (Inheritance)

User is the base class with common attributes (userID, name, email, password, phoneNumber) and methods (login (), logout ()).

Three specialized user types inherit from user:

- **Patient** – Books appointments, creates health posts, triggers SOS, consult AI.
- **Professional** – Conducts appointments, verifies posts.
  1. **Doctor**: General health, earns reward points and badges.
  2. **Phychologist**: Mental health specialist
- **Admin** – Verifies professionals, monitors payments, commissions and reports.

Supporting Classes

- **Health Post:** Patient posts verified by professionals.
- **Appointment**: Tracks date, time, fee and completion
- **Payment:** Holds payment until session completed.
- **Commission:** Platform earns a percentage of each appointment
- **Wallet:** Professional account for receiving payments.
- **Transaction**: Logs all financial movements
- **Emergency Contact**: Patient emergency contacts
- **SOS Alert**: Sends GPS location to contacts.
- **Subscription Plan**: Professional membership plans.
- **AI Service**: Recommends suitable professionals based on symptoms.

Workflows:

Appointment & Payment:

- Patient selects a professional and books an appointment.
- Patient pays the consultation fee
- Payment is help in escrow for security.
- Professional conducts the appointment/session.
- System calculates platform commission.
- Net amount is released to the professional wallet
- Payment and transaction records are updated

Health Post Verification:

- Patient creates a health post.
- Post status is marked as pending.
- Professional reviews and verifies the posts.
- Verified posts are published to the community.
- Professional earns reward points for verification.

Emergency SOS:

- Patient triggers SOS alert.
- System sends GPS location to all emergency contacts.
- Emergency contacts are notified instantly for assistance.

AI consultation:

- Patient describes symptoms in The AI system.
- AI analyzes the symptoms and recommends the best matching professional.

- Patient books an appointment directly with the recommended professional.

Relationships

- Patient => Health posts, Appointments, Emergency contacts, SOS alerts(1:M)
- Professional => Wallet (1:1)
- Professional => Appointment, Health posts(1:M)
- Appointment =>Payment (1:1)
- Payment => Transactions (1:M)
- Payment => Commission (1:1)
- Wallet => Transaction (1:M)

4.6 System Sequence Diagram
Sequence Diagram 1

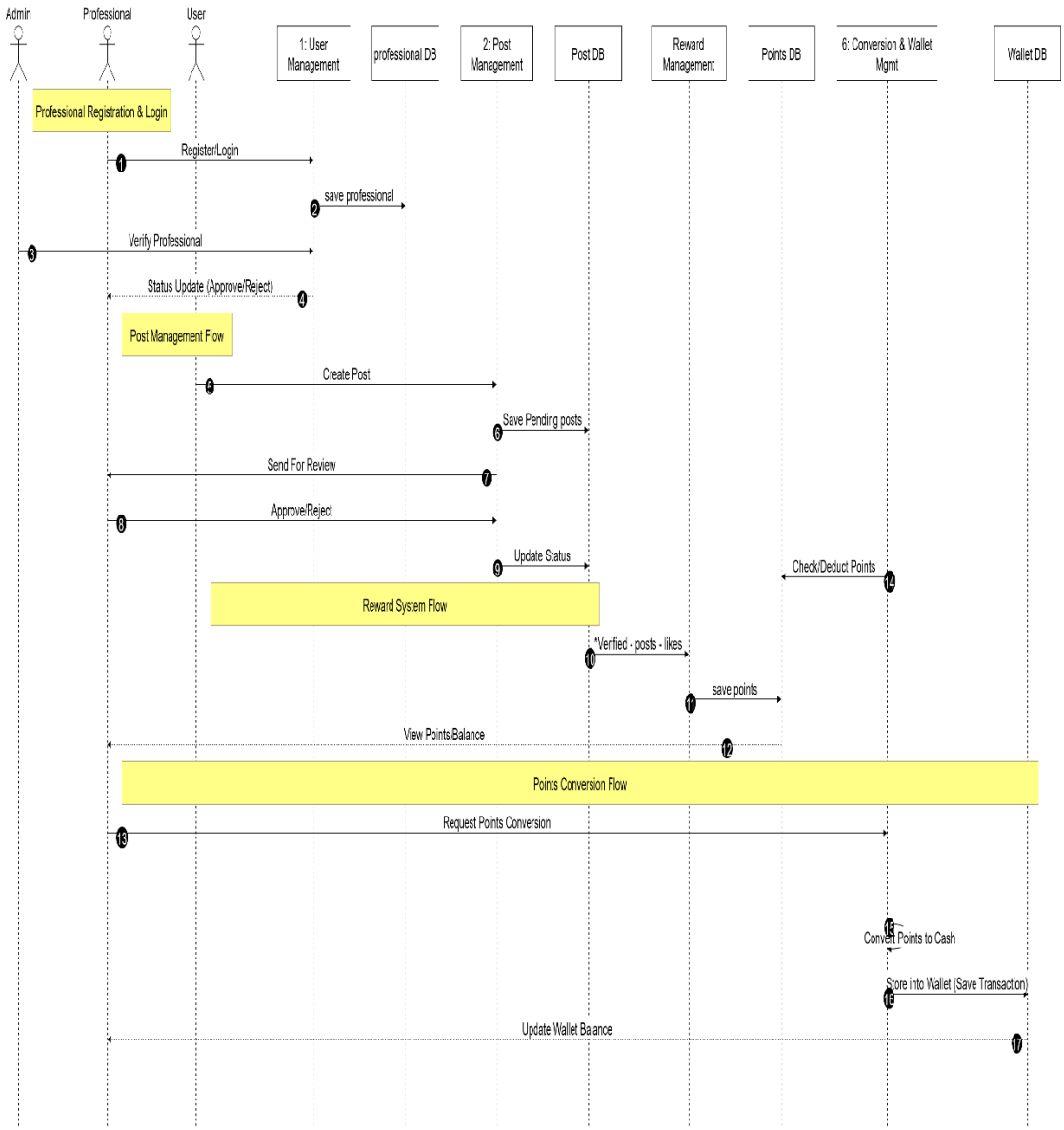

Figure 4.6 Health Post Creation and verification

This workflow shows how patients create health posts, how specialists verify them and how the chatbot assists patients.

Flow Description:

Use Case 1: Professional Registration and Verification

Step 1: Registration & Login

- Professional registers and logs into the system.

Step 2: Save professional data

- System stores professional details in the Professional Database.

Step 3: Admin verification

- Admin reviews professional information.

Step 4: Approval / Rejection

- Admin approves or rejects the professional and system updates status.

Use Case 2: Post Management

Step 5: Post Creation

- Patient click "New Post" and uploads media.
- Patient selects a category (e.g., skin, hair).

Step 6: Save for verification

- System saves the post with "Pending Verification" status to the database.

Step 7: Send post for verification

- System sends the post to the relevant medical professional for review.

Step 8: Approve or reject post

- Specialist reviews the post and either approves or rejects it.

If the post is approved:

Step 9: Status Update

- System updates the post status to "Verified" in the Database.

Use Case 3: Reward System Flow

Step 10: Verify Posts - likes

- System verifies the post likes to give point to specialist

Step 11: Save points

- System give point to specialist on behalf of verified posts-likes

Step 12: View points and balance

- Specialist view earned reward points.

Use Case 4: Points Conversion Flow

Step 13: Request Points Conversion

- Specialist request conversion of points to cash.

Step 14: Check and Deduct Points

- System verifies and deducts points.

Step 15: Convert Points into Cash

- Points are converted into cash

Step 16: Update Wallet and save transaction

- Converted Amount is added to wallet and transaction is stored.

Sequence Diagram 2

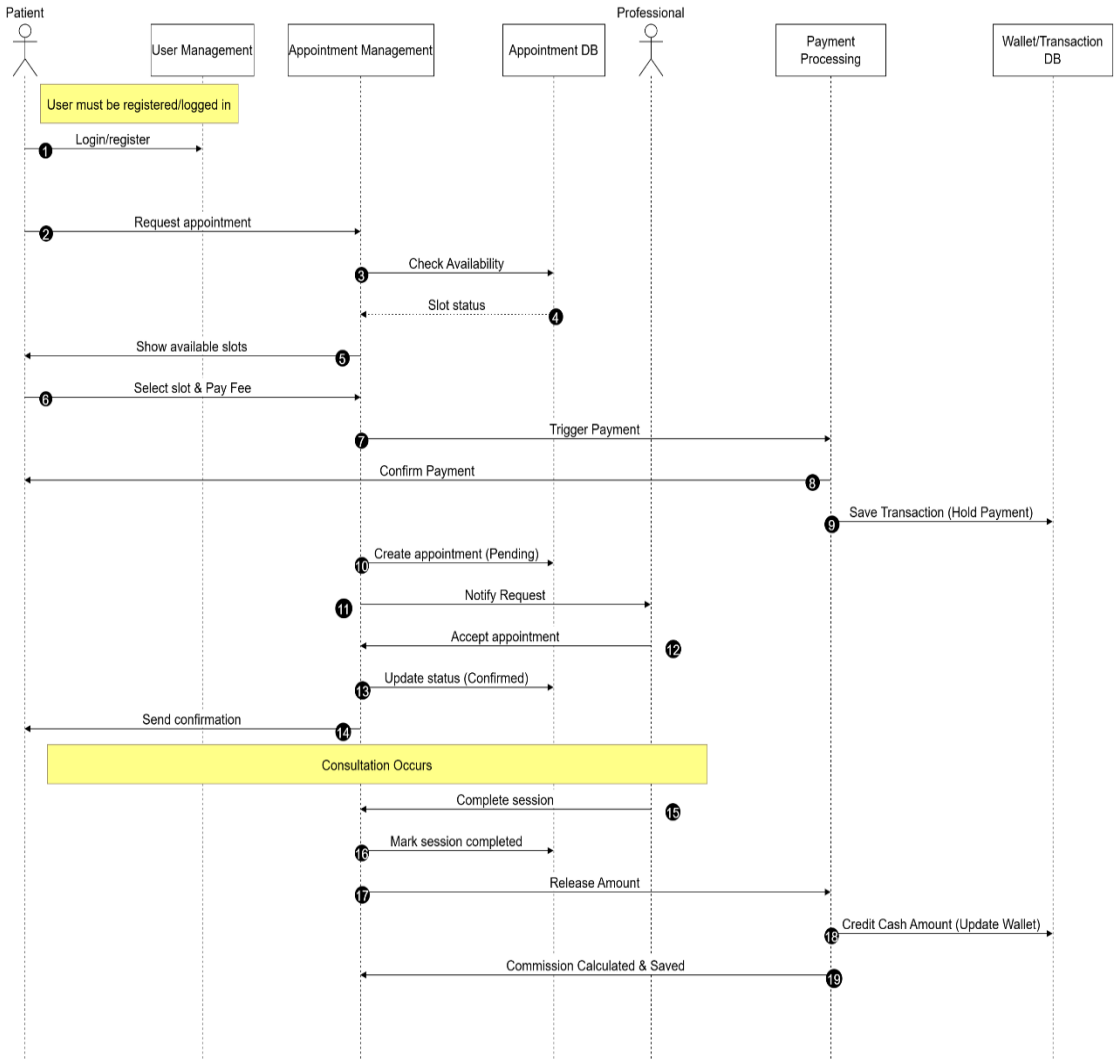

Figure 4.7 Book Appointment

The Book Appointment use case allows patients to schedule appointments with verified doctors through the system.

Flow Description:

Step 1: Login / Registration

- Patient logs in or registers in the system

Step 2: Appointment Request

- Patient sends a request to book an appointment.

Step 3: Availability Check

- System checks the professional's availability from the appointment database.

Step 4: Slot Status Response

- System receives slot status from the database.

Step 5: Slot Display

- Available time slots are shown to the patient.

Step 6: Slot Selection and fee payment

- Patient selects a slot and pays the fee.

Step 7: Trigger Payment

- System triggers the payment process through the payment service.

Step 8: Payment Confirmation

- Payment is triggered and confirmed.

Step 9: Hold Payment

- Transaction is saved as hold payment (escrow) in wallet/transaction DB.

Step 10: Appointment Creation

- Appointment is created with pending status.

Step 11: Professional Notification

- Professional receives appointment request notification.

Step 12: Appointment Acceptance

- Professional accepts the appointment request.

Step 13: Status update

- Appointment status is updated to confirmed.

Step 14: Patient Confirmation

- Confirmation message is sent to the patient.

Step 15: Consultation Session

- Online consultation between patient and professional takes place.

Step 16: Session Completion

- Professional makes the session as completed.

Step 17: Payment Release

- Held payment is released after session completion.

Step 18: Wallet Credit

- Remaining amount is credited to the professional's wallet.

Step 19: Commission Calculation

- Platform commission is calculated and recorded.

Sequence Diagram 3

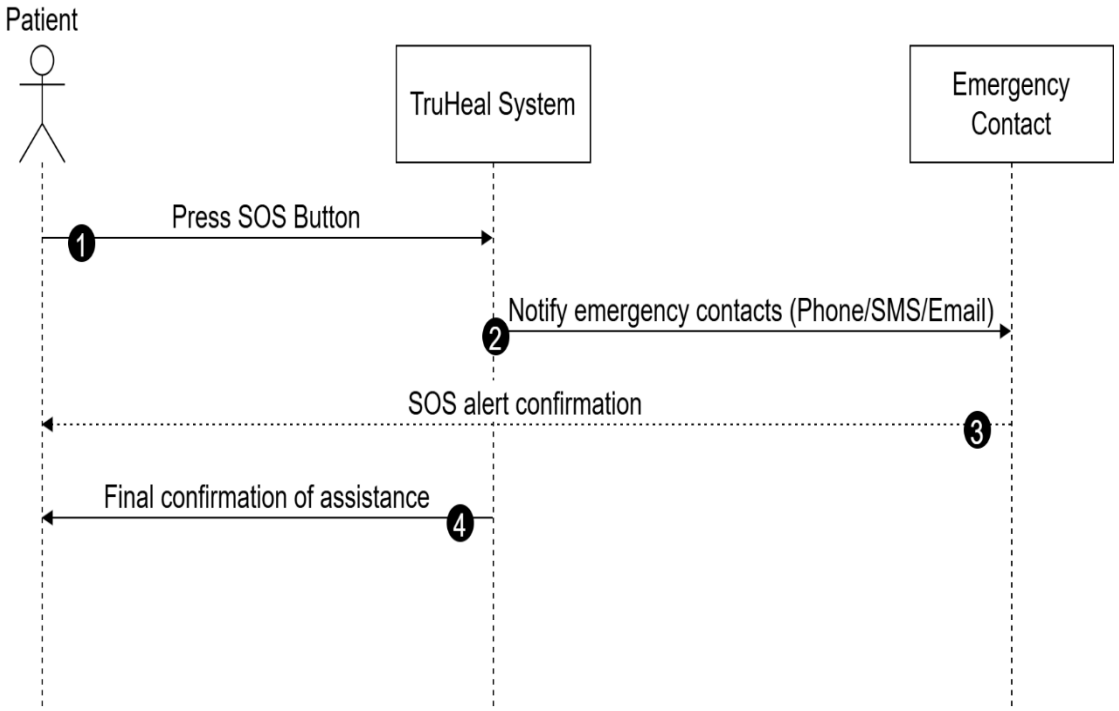

Figure 4.8 SOS Alert

This diagram shows how the SOS alert works step by step when a user presses the SOS button.

Flow Description:

Step 1: SOS Trigger

- User presses the SOS button in the TruHeal-Link system.

Step 2: Alert Notification

- The TruHeal System receives the request and immediately notifies all emergency contacts through phone call, SMS or email.

Step 3: Alert Confirmation

- Emergency contacts receive the SOS alert and send confirmation.

Step 4: Assistance Confirmation

- The system then sends a final confirmation of assistance back to the patient.

Purpose: This process ensures quick communication and immediate help during emergencies by instantly alerting the patients saved emergency contacts.

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